

Impact of Urbanization on Poverty in Vietnam

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May 17, 2026

Abstract

This study examines the relationship between urbanization and poverty in Vietnam and tests whether this relationship follows a non-linear inverted-U pattern. Nighttime light density is used as an objective proxy for urbanization, while poverty is measured using the Foster-Greer-Thorbecke indices: poverty incidence (P_0), poverty gap (P_1), and poverty severity (P_2). The analysis draws on a balanced household panel from the Vietnam Household Living Standards Survey in 2012, 2014, and 2016, combined with VIIRS satellite nightlight data aggregated at the district level. Fixed effects regressions are estimated at both district and household levels. The results show no statistically significant association between nightlight intensity and poverty at the district level. At the household level, the evidence is consistent with an inverted-U relationship only for the poverty gap. Years of schooling is the most consistent predictor of lower poverty, while ethnicity-related disparities remain important in some specifications.

Keywords: Urbanization, Poverty, Vietnam, Economic Development

JEL Codes: O18, I32, R11

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I. INTRODUCTION

Vietnam has experienced rapid structural transformation over the past two decades. The urban population share rose from 31.83 percent in 2012 to 34.44 percent in 2016, reflecting a sustained shift of labor from agriculture toward industry and services (General Statistics Office of Vietnam 2017). Over the same period, the national poverty rate fell from 16.8 to 5 percent between 2010 and 2020, lifting over ten million people out of poverty (World Bank 2022). Yet this aggregate progress hides large differences across groups: poverty remains concentrated among ethnic minority populations and remote rural districts even as urban centers continue to expand, and poverty rates range from 11.2 percent in the smallest towns to 1.9 percent in the largest cities (Ha et al. 2021). Whether urbanization genuinely drives poverty alleviation — and when its benefits start to reverse — remains an open empirical question with direct implications for Vietnam’s ambition to achieve high-middle-income status.

The existing literature has made progress on this question but leaves three gaps that this study addresses. Ha et al. (2021) and Nguyen et al. (2024) document negative and spatially differentiated urbanization effects on poverty at the provincial level, and Ha et al. (2021) estimate an urbanization threshold of approximately 40–44 percent beyond which poverty begins to rise. However, both studies rely on provincial aggregates and self-reported urbanization measures, so the household-level mechanisms behind this non-linearity are not identified. This study addresses these gaps along three dimensions. First, the analysis disaggregates to the district and household levels using three biennial waves of the VHLSS (2012–2016), instead of provincial aggregates used in prior work. Second, VIIRS satellite nighttime light data serve as an objective proxy for urbanization, which avoids the measurement error in self-reported urbanization rates. Third, the paper applies fixed effects specifications at both district and household levels. These specifications absorb time-invariant unobserved heterogeneity across districts and households, while province-year fixed effects account for province-specific shocks over time. Although this design does not fully eliminate endogeneity concerns, it provides more disciplined panel-based evidence than purely cross-sectional comparisons.

The results can be summarized as follows. At the district level, nighttime light intensity exhibits no statistically significant effect on any FGT poverty measure. At the household level, the estimates are consistent with an inverted-U association between nighttime light intensity and poverty depth (P_1), with an estimated turning point of approximately 52–62.5 nW/cm²/sr. This pattern suggests that early-stage urbanization may initially widen the poverty gap, while higher levels of urbanization may reduce it as labor markets and infrastructure become more developed. However, the relationship is not statistically significant for poverty incidence (P_0) or poverty severity (P_2), indicating that the poverty effects of urbanization are limited and dimension-specific. Years of schooling and ethnic minority status are consistent predictors of poverty across both levels of analysis. These results suggest that urbanization alone does not reduce poverty

in all dimensions. Education investment and support for ethnic minority households are necessary alongside urbanization policy.

The remainder of this paper is organized as follows. Section II presents the theoretical framework and hypothesis development. Section III describes the data sources, variable construction, and empirical methodology. Section IV reports and discusses the regression results at both the district and household levels. Section V summarizes the findings, discusses their policy implications, and outlines directions for future research.

II. HYPOTHESIS/THEORY/BACKGROUND

A. Theoretical Framework

Urbanization is more than migration to cities. It is a process in which a larger share of the population lives in urban areas, together with a structural shift of the economy from agriculture to industry and services. The key idea of urbanization lies in the concept of Structural Transformation Theory, which refers to the allocation of activities in agriculture, manufacturing, and services sectors ([Herrendorf et al. 2014](#)). Such a phenomenon can be explained by the Lewis model in which labor's marginal product in rural economies is zero or nearly zero, which indicates surplus labor. Consequently, surplus labor is hired at relatively low costs and wages do not increase because urban wages exceed the subsistence levels in rural economies. As a result, capitalists generate high profits, which are invested in capital formation. This process continues until the labor surplus is exhausted and the economy reaches the Lewis Turning Point, where rural wages rise until they equal urban wages ($\bar{w} = w_i$), halting further migration ([Lewis 1954](#)).

Nevertheless, the Harris-Todaro model argues that migration decisions are driven by expected wages rather than actual wages — a weighted average of formal and informal sector wages given the probability of formal employment. Because expected urban wages are usually higher than rural wages, migration continues even when urban unemployment exists. This leads to an informal sector and persistent urban poverty ([Harris and Todaro 1970](#)).

B. The Global Definitions of Poverty

The definition of poverty has many forms because researchers in different fields study it differently ([Gordon 2005](#)). A common emphasis is that long periods of low income limit opportunities to improve living standards. However, the generalization of absolute poverty defined by the United Nations (UN) in 1995 fully captured this abstract concept. Overall, poverty—while not restricted to earnings—is characterized by a lack of income and adequate resources to sustain a basket of basic human needs. This includes food, shelter, hydration, sanitation facilities, education, information, and access to other essential services. Beyond depending on income levels, the concept of poverty can also extend to the deprivation of participation in decision-making, civil, cultural, and social life of the poor.

C. The Measurement of Poverty: Foster-Greer-Thorbecke (FGT)

The Foster-Greer-Thorbecke (FGT) index is the standard tool for measuring poverty in the empirical literature. The measurement, while aligning with the definition, is defined by the sum of normalized income shortfalls to the power of alpha and divided by the total population. When alpha equals 0, it becomes the headcount index, which

is the ratio between poor households and the total population. As a result, it can be interpreted as the rate of poverty or poverty incidence (P_0). When alpha is 1, the formula is shortened to the headcount multiplied by the income gap ratio, which is understood as the poverty gap and depth (P_1). When alpha equals 2, it captures the squared distance to the poverty line, also known as poverty severity (P_2).

D. Historical Research and Background

Fan et al. (2018) examine socio-economic and environmental changes under Doi Moi, measuring urban land density via nighttime light and showing trade-offs between economic growth and environmental quality. Poverty in this literature was measured as a headcount rate, an approach that remained appropriate prior to Vietnam's introduction of the multidimensional poverty rate in 2015.

Ha et al. (2021) and [Nguyen et al. \(2024\)](#) further document the spatial and spill-over effects of urbanization through the provincial and cities dataset. Unlike [Ha et al. \(2021\)](#), [Nguyen et al. \(2024\)](#) find a consistently negative effect of urbanization on poverty over the period 2016–2023. In addition, this effect declines with distance: as the number of surrounding urban residents increases, poverty in neighboring areas decreases, with the magnitude diminishing as distance grows.

E. Non-linear Relationship and Urbanization of Poverty

Several studies also show that the relationship between urbanization and poverty is non-linear. The Kuznets hypothesis and its spatial extensions suggest that urbanization affects poverty in two stages. In the early stage, urban growth can widen inequality. After a certain point, the benefits spread to lower-income groups. This pattern depends on urban carrying capacity. When cities can still absorb rural labor, urbanization reduces poverty.

However, when urbanization grows faster than infrastructure and institutions can support, the costs start to exceed the benefits. These costs include congestion, high housing prices, informal settlements, pollution, and a larger gap between urban and rural wages. Poor migrants often get trapped in these conditions. Following [Ravallion et al. \(2007\)](#) and [Martinez-Vazquez et al. \(2009\)](#), excessive urbanization triggers the “urbanization of poverty”, where the rural poor migrate to urban centers but stay excluded from the formal economy.

F. Household Heterogeneity and Control Variables

Education and urbanization are correlated because skilled workers tend to move to cities ([Hanushek and Woessmann 2008](#); [Psacharopoulos and Patrinos 2018](#)). If we do not control for education, the estimated urbanization effect will absorb the return to schooling. We therefore include average years of schooling in the regression to separate these two effects.

We also control for the household dependency ratio—the proportion of non-working-age members relative to working-age adults. A higher dependency ratio lowers per capita consumption and reduces labor force participation, especially among women, which limits household income (White and Masset 2003; Eastwood and Lipton 1999).

These theories suggest the relationship between urbanization and poverty is unclear. On one hand, urbanization can reduce poverty through structural change, higher-productivity jobs, better infrastructure, and improved public services. On the other hand, when migration exceeds what urban labor and housing markets can absorb, urbanization may increase poverty through informal jobs, higher living costs, congestion, and exclusion of low-skill workers. This ambiguity motivates the empirical test of both a linear relationship and a non-linear inverted-U pattern.

G. Hypothesis

To examine the determinants of poverty at the district and household level, the following hypotheses are proposed:

- **H1:** Conditional on district, household, and province-year fixed effects, nighttime light intensity is significantly associated with poverty.
- **H2:** The relationship between nighttime light intensity and poverty is non-linear and follows an inverted-U pattern, where poverty initially increases with urbanization but declines after a threshold level of urbanization.

In this study, nighttime light intensity serves as an indicator of urbanization because higher nighttime light reflects greater economic activity, infrastructure development, and population concentration in urban areas. We also hypothesize an inverted-U relationship: early-stage migrants face welfare losses at first, then earn higher incomes after staying longer in urban labor markets. Three control variables are included in the regression analysis: average years of schooling, household dependency ratio, and ethnic minority ratio.

III. DATA AND METHODOLOGY

In this study, we utilize a balanced panel dataset spanning three biennial waves of the Vietnam Household Living Standards Survey (VHLSS) in 2012, 2014, and 2016, a nationally representative survey conducted by the General Statistics Office (GSO) of Vietnam. We restrict the sample to households observed in all three waves to reduce composition bias. We pair this micro-data with high-resolution satellite imagery from the Visible Infrared Imaging Radiometer Suite (VIIRS) to capture spatial variation.

A. Variable Selection

1. Dependent Variables: FGT Poverty Indices (P_0 , P_1 , P_2)

We measure poverty incidence (P_0), depth (P_1), and severity (P_2). Using per capita real total household expenditure adjusted via the spatial cost of living index (SCOLI), welfare is evaluated against the GSO-World Bank cost of basic needs (CBN) poverty lines.

2. Independent Variable: Nighttime Light Density

We use nighttime light density as an objective spatial proxy for urbanization, which avoids the measurement error in self-reported data. The data come from VIIRS Annual V2.1 (median masked). We cropped and masked the global rasters to Vietnam's administrative boundaries. We compute the mean nighttime intensity (urbanization index) via geospatial zonal statistics. Household-level nighttime values are assigned based on the mean nighttime intensity of the district in which the household resides.

3. Control Variables

Proportion of Dependency

In VHLSS, some surveyed households do not have working-age people (i.e., elderly living alone, or children without working-age adults); thus to prevent the division by zero in the proportion of dependency, we assign a dependency proportion of one. The proportion of dependency is then calculated as follows:

$$\text{Proportion of dependency} = \frac{\text{Dependent Population}}{\text{Total Population}}$$

Based on the definition of working age in Vietnam, which is above 15 years old, the working-age population is categorized as male no larger than 60, and female no larger than 55. Therefore, the dependency group is those outside the threshold (i.e., < 15 or > 55 or > 60).

Average Years of Schooling

The average years of schooling is a proxy for the education parameter. This is calculated

as the mean of the highest grade completed across all members of the household or all individuals within the district who have education data.

Ethnicity Ratio

The ethnicity ratio measures the proportion of ethnic minorities within the district. At the household level, the ethnic variable is considered as a dummy variable with 1 standing for Kinh people, and 0 for other minority groups.

Table 1
SUMMARY STATISTICS FOR DISTRICT-LEVEL MODEL

Variable	Obs	Mean	Std. Dev.	Min	Max
Poverty Incidence (P_0)	2,068	0.163	0.234	0.000	1.000
Poverty Gap (P_1)	2,068	0.046	0.089	0.000	0.737
Poverty Severity (P_2)	2,068	0.019	0.047	0.000	0.562
Mean Nightlight	2,064	2.605	8.174	0.000	108.389
Nightlight squared	2,064	73.565	470.885	0.000	11,748.060
Ethnicity ratio	2,068	0.233	0.354	0.000	1.000
Proportion of dependency	2,068	0.377	0.084	0.000	0.778
Years of schooling	2,068	6.449	1.561	1.826	10.727

Notes: Summary statistics are calculated based on the district-level balanced panel.

Table 2
SUMMARY STATISTICS FOR HOUSEHOLD-LEVEL MODEL

Variable	Obs	Mean	Std. Dev.	Min	Max
Poverty Incidence (P_0)	5,685	0.129	0.335	0.000	1.000
Poverty Gap (P_1)	5,685	0.033	0.106	0.000	0.796
Poverty Severity (P_2)	5,685	0.012	0.052	0.000	0.634
Mean Nightlight	5,592	2.832	7.761	0.000	76.803
Nightlight squared	5,592	68.237	359.302	0.000	5,898.672
Ethnicity (Kinh=1)	5,685	0.826	0.379	0.000	1.000
Proportion of dependency	5,685	0.390	0.304	0.000	1.000
Years of schooling	5,685	6.654	2.869	0.000	12.000

Notes: Summary statistics are calculated based on the household-level balanced panel.

B. Model Specification

We use a fixed effects model to estimate the impact of urbanization on poverty in panel data. The FE estimator removes time-invariant differences across spatial units by demeaning each variable over time within each unit. As a result, any unobservable factors that do not change over time are excluded from the estimation process.

Fixed effects models using `reghdfe` are estimated separately at the district and household levels across the 2012, 2014, and 2016 waves, with the squared nighttime light term included to capture potential non-linearity.

1. District Level Model

$$\text{poverty}_{dt} = \beta_0 + \beta_1 \text{nightlight}_{dt} + \beta_2 \text{nightlight}_{dt}^2 + \sum_{k=3}^n \beta_k X_{dtk} + \alpha_d + \delta_{pt} + u_{dt}$$

Where:

- poverty_{dt} : poverty indicators (P_0, P_1, P_2) for district d at time t
- nightlight_{dt} : mean nighttime light in district d at time t
- nightlight_{dt}^2 : the squared term of mean nighttime light
- X_{dt} : control variables (schooling, ethnicity ratio, proportion of dependency)
- α_d : district-level fixed effects
- δ_{pt} : province-year fixed effects
- u_{dt} : idiosyncratic error term

2. Household Model

$$\text{poverty}_{hdt} = \beta_0 + \beta_1 \text{nightlight}_{dt} + \beta_2 \text{nightlight}_{dt}^2 + \sum_{k=3}^n \beta_k X_{hdtk} + \alpha_d + \gamma_h + \delta_{pt} + u_{hdt}$$

Where:

- poverty_{hdt} : poverty indicators (P_0, P_1, P_2) for household h in district d at time t
- γ_h : household-level fixed effects
- X_{hdt} : household-level control variables

In other words, the fixed effects model can control for spatial effects, which in this case are district-level fixed effects (α_d) and household-level fixed effects (γ_h), since there are unobserved differences across districts and households that need to be accounted for. Moreover, the FE model can also control for temporal effects by including year fixed effects, which capture time-specific factors affecting all units. Normally, dummy year variables are created and added to the model to represent year fixed effects; however, year effects are instead incorporated in the model through province-year interaction fixed effects in this study, which capture time-varying shocks at the provincial level.

IV. RESULTS

A. Estimators and Statistical Significance

Table 3
DISTRICT-LEVEL REGRESSION RESULTS ON POVERTY INDICES

VARIABLES	Poverty Incidence (P_0)	Poverty Gap (P_1)	Poverty Severity (P_2)
	(1)	(2)	(3)
Mean Nightlight	0.0006 (0.0052)	-0.0003 (0.0012)	-0.0002 (0.0005)
Nightlight squared	-0.00002 (0.00003)	-0.00000 (0.00001)	0.00000 (0.00000)
Ethnicity ratio	0.2127 (0.0911)	0.0869 (0.0465)	0.0454 (0.0274)
Proportion of dependency	-0.0440 (0.0816)	0.0134 (0.0336)	0.0094 (0.0207)
Years of schooling	-0.0538 (0.0086)	-0.0141 (0.0036)	-0.0053 (0.0020)
Observations	2,034	2,034	2,034
R-squared	0.871	0.860	0.837
Adj R-squared	0.780	0.760	0.720
District FE	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes
VCE Cluster	Province	Province	Province

Notes: All models are estimated at the district level using `reghdfe` on a balanced panel.

Robust standard errors clustered at the province level (62 clusters) in parentheses.

After running the regression at the district level, the outputs show that mean nightlight and mean nightlight squared are not statistically significant under any level for all three poverty measures. To make magnitudes easier to interpret, we use one-standard-deviation changes as the main unit of analysis. Among the three poverty measures, one-standard-deviation increases in nightlight yield mixed directional effects across P_0 , P_1 , and P_2 , though none reach statistical significance. The positive net marginal effect on P_1 evaluated at mean nightlight levels comes from within-sample averaging, not from the turning-point pattern discussed at the household level. Among control variables, ethnicity ratio and years of schooling are statistically significant at the 1% level across all three poverty measures. Although the constants are not the concern of the study, they remain statistically significant at the 1% level across all measures. The model fit is strong, with R-squared ranging from 0.837 to 0.871 across specifications.

Table 4
HOUSEHOLD-LEVEL REGRESSION RESULTS ON POVERTY INDICES

VARIABLES	Poverty Incidence (P_0)		Poverty Gap (P_1)		Poverty Severity (P_2)	
	(1)	(2)	(3)	(4)	(5)	(6)
Mean Nightlight	0.0081 (0.0059)	0.0083 (0.0059)	0.0025 (0.0012)	0.0024 (0.0012)	0.0010 (0.0006)	0.0009 (0.0006)
Nightlight squared	-0.00008 (0.00005)	-0.00008 (0.00006)	-0.00002 (0.00001)	-0.00002 (0.00001)	-0.00001 (0.00000)	-0.00001 (0.00000)
Ethnicity (Kinh=1)	0.0077 (0.0042)	-0.0067 (0.0052)	0.0044 (0.0020)	0.0001 (0.0014)	0.0022 (0.0012)	0.0001 (0.0006)
Proportion of Dependency	0.0137 (0.0171)	-0.0100 (0.0262)	0.0094 (0.0056)	0.0162 (0.0068)	0.0048 (0.0028)	0.0108 (0.0032)
Years of schooling	-0.0226 (0.0025)	-0.0173 (0.0036)	-0.0049 (0.0008)	-0.0021 (0.0009)	-0.0017 (0.0004)	-0.0003 (0.0004)
Observations	5,592	5,592	5,592	5,592	5,592	5,592
R-squared	0.495	0.692	0.572	0.752	0.546	0.720
Adj R-squared	0.4177	0.4243	0.5067	0.5363	0.4762	0.4754
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Province-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	No	Yes	No	Yes	No	Yes
VCE Cluster	District	District	District	District	District	District

Notes: All models are estimated at the household level using `reghdfe` on a balanced panel.

Province-Year FE refers to the interaction between province and year dummies. Robust standard errors clustered at the district level in parentheses.

At the household level, the outputs show that both mean nightlight and mean nightlight squared are only statistically significant at 5% level under the P_1 regression model with or without household-level fixed effects. In terms of magnitudes, one-standard-deviation increases in nightlight are associated with substantial percentage changes in all three poverty measures at the household level, though only the P_1 effect reaches statistical significance. The average marginal effect on P_1 is 0.0025 per unit increase in nightlight (0.0023 in the within-household model). The constants are statistically significant across all three poverty measures. The household proportion of dependency is significant at the 1% level for P_2 and at the 5% level for P_1 , but insignificant for P_0 . Years of schooling exhibit a significant effect on P_0 at the 1% level and on P_1 at the 5% level, but are statistically insignificant for P_2 . The adjusted R-squared in the within-household models (columns 2, 4, 6) is lower than in the between-district models, reflecting the limited within-unit variation over the short panel.

B. District Level Analysis

At the district level, neither mean nighttime light nor its squared term is statistically significant for any of the three FGT poverty measures. Therefore, the district-level results do not provide evidence that changes in urbanization, as proxied by nighttime

light intensity, are systematically associated with changes in poverty during the 2012–2016 period. The signs of the coefficients differ across poverty measures, but given the large standard errors and insignificant p-values, these patterns should not be interpreted as robust evidence of either poverty reduction or poverty increase.

Beyond the primary effect, the control variables provide critical insights into demographic aspects for poverty research. With a strongly significant positive correlation between the ethnicity ratio and poverty, the disparities between ethnic groups are highlighted as the underlying cause for local poverty status. In addition, average years of schooling, which proxies for human capital, has a strong negative effect on poverty across all three dimensions (incidence, depth, and severity). This suggests that education investment is important for reducing poverty.

These findings suggest that urbanization alone is not enough to reduce poverty. For long-term development, investment in human capital and support for disadvantaged groups are necessary.

With respect to the hypotheses proposed in Section G, the district-level results do not support H1: urbanization does not exhibit a statistically significant effect on poverty at this level of aggregation. Therefore, H2 — the inverted-U relationship between nightlight intensity and poverty — cannot be confirmed at the district level either, because both the linear and quadratic nightlight terms are not significant, so we cannot identify a turning point. These null results do not mean that urbanization does not matter for poverty. They more likely reflect the limited within-district variation in our short three-wave panel, as we discuss in the household-level analysis below.

C. Household-Level Analysis

At the household level, H1 receives partial support. Mean nightlight and its squared term are statistically significant at the 5 percent level exclusively under the P_1 specification in both the between-district and within-household models ($p = 0.040$ and 0.048 respectively; Table A2), while no significance is detected for P_0 or P_2 under either specification. The fact that only P_1 is significant is consistent with how the three FGT measures work. P_0 only captures whether a household is above or below the poverty line, so small income changes do not show up. P_2 puts most weight on the poorest households, whose poverty is structural and unlikely to change with urbanization in a few years. P_1 measures the average income gap to the poverty line, so it responds more to the kind of marginal income changes that urbanization tends to produce.

The positive linear and negative quadratic coefficients on nightlight for P_1 are consistent with the inverted-U pattern in H2 and match the Harris-Todaro mechanism in Section A: at low nightlight levels, migrants bear the costs of moving to urban areas — informal jobs, higher living costs — before urban labor markets give them more stable formal jobs at higher urbanization levels. The estimated turning point of $NL^* \approx 52\text{--}62.5 \text{ nW/cm}^2/\text{sr}$ ¹ falls within the observed household-level nightlight distribution

¹The turning point $NL = -\beta_1/(2\beta_2)$ is computed using full-precision (unrounded) coefficients;

(maximum $76.8 \text{ nW/cm}^2/\text{sr}$), which means the curvature is within our sample. The marginal effect schedules in Tables A4 and A5 quantify this trajectory: at $NL = 0$, a 0.1 standard deviation increase in nightlight is associated with a 5.914 percentage point increase in P_1 . This is consistent with Harris-Todaro: early migrants bear costs before finding stable formal jobs. This positive effect declines monotonically, turning negative at approximately $NL^* \approx 52 \text{ nW/cm}^2/\text{sr}$ and reaching -1.884 percentage points at $NL = 70$ in the between-district model (Table A4; Figure B6). The within-household model yields a closely comparable pattern (Table A5; Figure B6), which suggests the result is consistent across specifications..

Mean nightlight loses statistical significance after we add household fixed effects. This is because nightlight is a district-level variable with limited within-household variation over three biennial waves, so the identification becomes less precise. The poverty-reducing effect of urbanization is therefore more a cross-sectional pattern than a within-household change over time.

Among control variables, years of schooling is negative and significant at the 1 percent level for P_0 and at the 5 percent level for P_1 in the between-district model, but insignificant for P_2 . The ethnicity dummy is significant for P_0 and P_1 between districts but loses significance under household fixed effects. The dependency ratio is significant at the 1 percent level for P_2 and at the 5 percent level for P_1 in the within-household model.

rounding in Table 4 accounts for the apparent discrepancy.

V. CONCLUSION

A. Summary

This study examines the relationship between urbanization and poverty in Vietnam using a balanced household panel from VHLSS 2012, 2014, and 2016, combined with VIIRS nighttime light data. Poverty is measured using the FGT indices (P_0 , P_1 , and P_2), and urbanization is proxied by district-level nighttime light intensity.

At the district level, nighttime light intensity has no statistically significant effect on any of the three FGT poverty measures. This suggests that within-district changes in urbanization over the short panel are too small to produce detectable poverty reductions. At the household level, urbanization has a statistically significant inverted-U effect only on poverty depth (P_1), with an estimated turning point of approximately 52–62.5 nW/cm²/sr. This pattern is consistent with the Harris-Todaro framework: early migrants face welfare losses before urban labor markets give them more stable jobs. The effect on poverty incidence (P_0) and poverty severity (P_2) does not reach statistical significance. This reflects two things: P_0 is a binary measure that misses smaller welfare changes, while P_2 puts most weight on chronically poor households whose poverty is harder to change in a short period.

Among the control variables, years of schooling has a significant negative effect on poverty in both models. This confirms that education is an important factor in poverty reduction. Ethnic minority status is associated with higher poverty at the district level and in the between-district household model, which shows that non-Kinh households still face structural disadvantages. The dependency ratio is significantly associated with poverty severity (P_2) in the within-household model, consistent with the resource-dilution hypothesis.

Overall, these results suggest that urban expansion alone is not enough to reduce poverty in all dimensions. Education investment and support for ethnic minority households are necessary alongside urbanization policy if Vietnam wants to reach high-middle-income status.

B. Limitations

Several limitations should be noted. First, the estimates rely on self-reported survey data (VHLSS), which may be subject to social desirability bias. Additionally, nightlight data is aggregated to the district level using 2022 administrative boundaries. Due to data availability constraints, district boundaries are defined according to this 2022 delineation. Administrative reorganizations occurring between 2012 and 2022 — such as district mergers or splits — may introduce misclassification in the spatial assignment of nightlight values, which could weaken the estimated urbanization effects. Additionally, the study does not fully address potential endogeneity: nighttime light density may itself be driven by poverty levels (reverse causality), and omitted variables correlated with both urbanization and poverty cannot be entirely ruled out. Specifi-

cally, if districts receiving infrastructure investment simultaneously experience both rising nightlight and falling poverty, the FE estimator remains consistent. However, if transitory economic shocks simultaneously drive nightlight and poverty in opposite directions within the panel window, the estimated coefficients are biased toward zero. This means our null results at the district level may reflect attenuation rather than a true zero effect. Future research would benefit from longer panel data or the use of instrumental variables to address these identification challenges.

AI DECLARATION

In this work, we used Gemini in order to find the LaTeX code for formatting purpose. Also, since English is not our first language, we utilize Gemini for grammar correction.

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Appendix A: Tables

Table A1
P-VALUES FOR DISTRICT-LEVEL REGRESSION RESULTS

VARIABLES	Poverty Incidence (P_0)	Poverty Gap (P_1)	Poverty Severity (P_2)
	(1)	(2)	(3)
Mean Nightlight	0.914	0.789	0.655
Nightlight squared	0.507	0.843	0.953
Ethnicity ratio	0.023	0.066	0.103
Dependency ratio	0.592	0.692	0.651
Years of schooling	0.000	0.000	0.010
Observations	2,034	2,034	2,034

Notes: This table reports the p-values corresponding to the regression coefficients in Table 3.

Table A2
P-VALUES FOR HOUSEHOLD-LEVEL REGRESSION RESULTS

VARIABLES	Poverty Incidence (P_0)		Poverty Gap (P_1)		Poverty Severity (P_2)	
	(1)	(2)	(3)	(4)	(5)	(6)
Mean Nightlight	0.170	0.161	0.040	0.048	0.090	0.117
Nightlight squared	0.150	0.166	0.022	0.047	0.067	0.148
Ethnicity (Kinh=1)	0.064	0.199	0.025	0.939	0.061	0.843
Dependency ratio	0.422	0.703	0.093	0.017	0.082	0.001
Years of schooling	0.000	0.000	0.000	0.017	0.000	0.441
Model	Between	Within	Between	Within	Between	Within
Observations	5,592	5,592	5,592	5,592	5,592	5,592

Notes: This table reports the p-values corresponding to the regression coefficients in Table 4.

"Between" refers to the Between-District model; "Within" refers to the Within-Household model.

Table A3
MARGINAL EFFECTS OF URBANIZATION ON POVERTY (DISTRICT LEVEL)

NL Intensity	Poverty Incidence (P_0)		Poverty Gap (P_1)		Poverty Severity (P_2)	
	0.1 SD	1 SD	0.1 SD	1 SD	0.1 SD	1 SD
0	0.281%	2.813%	-0.580%	-5.805%	-1.043%	-10.427%
	[-4.810, 5.373]	[-48.100, 53.726]	[-4.815, 3.654]	[-48.151, 36.541]	[-5.595, 3.510]	[-55.952, 35.098]
10	0.050%	0.496%	-0.637%	-6.365%	-1.026%	-10.259%
	[-4.390, 4.490]	[-43.904, 44.895]	[-4.358, 3.085]	[-43.581, 30.851]	[-5.068, 3.017]	[-50.684, 30.166]
20	-0.182%	-1.821%	-0.693%	-6.926%	-1.009%	-10.091%
	[-3.981, 3.617]	[-39.811, 36.168]	[-3.914, 2.529]	[-39.141, 25.290]	[-4.555, 2.537]	[-45.550, 25.369]
30	-0.414%	-4.139%	-0.749%	-7.486%	-0.992%	-9.923%
	[-3.588, 2.761]	[-35.885, 27.608]	[-3.490, 1.993]	[-34.902, 19.930]	[-4.062, 2.077]	[-40.618, 20.772]
40	-0.646%	-6.456%	-0.805%	-8.046%	-0.975%	-9.755%
	[-3.225, 1.933]	[-32.246, 19.335]	[-3.099, 1.490]	[-30.988, 14.895]	[-3.599, 1.649]	[-35.995, 16.485]
50	-0.877%	-8.773%	-0.861%	-8.607%	-0.959%	-9.587%
	[-2.915, 1.160]	[-29.149, 11.603]	[-2.763, 1.042]	[-27.632, 10.419]	[-3.187, 1.269]	[-31.869, 12.695]
60	-1.109%	-11.090%	-0.917%	-9.167%	-0.942%	-9.419%
	[-2.715, 0.497]	[-27.150, 4.970]	[-2.525, 0.691]	[-25.247, 6.912]	[-2.855, 0.971]	[-28.550, 9.712]
70	-1.341%	-13.408%	-0.973%	-9.728%	-0.925%	-9.251%
	[-2.731, 0.050]	[-27.314, 0.499]	[-2.443, 0.497]	[-24.427, 4.972]	[-2.649, 0.799]	[-26.487, 7.985]
80	-1.572%	-15.725%	-1.029%	-10.288%	-0.908%	-9.083%
	[-3.061, -0.084]	[-30.609, -0.840]	[-2.560, 0.503]	[-25.603, 5.027]	[-2.611, 0.794]	[-26.105, 7.939]
90	-1.804%	-18.042%	-1.085%	-10.848%	-0.892%	-8.915%
	[-3.655, 0.046]	[-36.547, 0.463]	[-2.857, 0.687]	[-28.567, 6.870]	[-2.746, 0.963]	[-27.462, 9.632]
100	-2.036%	-20.359%	-1.141%	-11.409%	-0.875%	-8.747%
	[-4.394, 0.322]	[-43.939, 3.221]	[-3.272, 0.991]	[-32.723, 9.905]	[-3.019, 1.270]	[-30.190, 12.696]

Note: Percentage changes in poverty indices for a 0.1 and 1 standard deviation increase in nightlight.
95% confidence intervals in brackets.

Table A4
**MARGINAL EFFECTS OF URBANIZATION ON POVERTY (HOUSEHOLD LEVEL -
 BETWEEN-DISTRICT)**

NL Intensity	Poverty Incidence (P_0)		Poverty Gap (P_1)		Poverty Severity (P_2)	
	0.1 SD	1 SD	0.1 SD	1 SD	0.1 SD	1 SD
0	4.822%	48.220%	5.914%	59.141%	6.133%	61.333%
	[-2.057, 11.702]	[-20.575, 117.015]	[0.281, 11.547]	[2.809, 115.472]	[-0.936, 13.202]	[-9.357, 132.023]
10	3.876%	38.763%	4.800%	48.001%	5.046%	50.463%
	[-1.781, 9.534]	[-17.810, 95.336]	[0.076, 9.524]	[0.758, 95.244]	[-0.905, 10.997]	[-9.047, 109.973]
20	2.931%	29.305%	3.686%	36.861%	3.959%	39.593%
	[-1.540, 7.401]	[-15.401, 74.012]	[-0.148, 7.521]	[-1.483, 75.205]	[-0.894, 8.812]	[-8.936, 88.121]
30	1.985%	19.848%	2.572%	25.722%	2.872%	28.722%
	[-1.373, 5.342]	[-13.727, 53.422]	[-0.408, 5.553]	[-4.084, 55.527]	[-0.920, 6.664]	[-9.195, 66.640]
40	1.039%	10.390%	1.458%	14.582%	1.785%	17.852%
	[-1.382, 3.460]	[-13.823, 34.604]	[-0.747, 3.663]	[-7.470, 36.634]	[-1.025, 4.595]	[-10.248, 45.952]
50	0.093%	0.933%	0.344%	3.442%	0.698%	6.982%
	[-1.845, 2.032]	[-18.452, 20.318]	[-1.281, 1.969]	[-12.805, 19.689]	[-1.328, 2.725]	[-13.283, 27.247]
60	-0.852%	-8.524%	-0.770%	-7.697%	-0.389%	-3.888%
	[-3.079, 1.374]	[-30.788, 13.739]	[-2.258, 0.718]	[-22.577, 7.182]	[-2.123, 1.346]	[-21.234, 13.458]
70	-1.798%	-17.982%	-1.884%	-18.837%	-1.476%	-14.758%
	[-4.874, 1.278]	[-48.739, 12.775]	[-3.777, 0.010]	[-37.774, 0.100]	[-3.621, 0.670]	[-36.212, 6.695]

Note: Results for Between-District model. Percentage changes in poverty indices for a 0.1 and 1 standard deviation increase in nightlight. 95% confidence intervals in brackets.

Table A5
MARGINAL EFFECTS OF URBANIZATION ON POVERTY (HOUSEHOLD LEVEL -
WITHIN-HOUSEHOLD)

NL Intensity	Poverty Incidence (P_0)		Poverty Gap (P_1)		Poverty Severity (P_2)	
	0.1 SD	1 SD	0.1 SD	1 SD	0.1 SD	1 SD
0	4.951%	49.506%	5.779%	57.790%	5.794%	57.936%
	[-1.961, 11.862]	[-19.613, 118.625]	[0.058, 11.499]	[0.585, 114.995]	[-1.449, 13.036]	[-14.489, 130.360]
10	4.031%	40.307%	4.793%	47.932%	4.900%	49.002%
	[-1.647, 9.708]	[-16.469, 97.082]	[0.006, 9.580]	[0.065, 95.799]	[-1.177, 10.977]	[-11.767, 109.771]
20	3.111%	31.107%	3.807%	38.074%	4.007%	40.068%
	[-1.370, 7.592]	[-13.704, 75.918]	[-0.064, 7.679]	[-0.637, 76.786]	[-0.926, 8.939]	[-9.257, 89.392]
30	2.191%	21.908%	2.822%	28.217%	3.113%	31.134%
	[-1.172, 5.554]	[-11.725, 55.540]	[-0.169, 5.812]	[-1.688, 58.121]	[-0.715, 6.942]	[-7.149, 69.417]
40	1.271%	12.709%	1.836%	18.359%	2.220%	22.200%
	[-1.164, 3.706]	[-11.640, 37.057]	[-0.351, 4.023]	[-3.514, 40.232]	[-0.592, 5.032]	[-5.921, 50.320]
50	0.351%	3.509%	0.850%	8.501%	1.327%	13.266%
	[-1.632, 2.334]	[-16.317, 23.336]	[-0.734, 2.434]	[-7.341, 24.344]	[-0.695, 3.348]	[-6.945, 33.477]
60	-0.569%	-5.690%	-0.136%	-1.356%	0.433%	4.332%
	[-2.875, 1.737]	[-28.753, 17.373]	[-1.592, 1.320]	[-15.915, 13.203]	[-1.352, 2.219]	[-13.523, 22.187]
70	-1.489%	-14.889%	-1.121%	-11.214%	-0.460%	-4.602%
	[-4.666, 1.688]	[-46.659, 16.880]	[-3.022, 0.780]	[-30.223, 7.795]	[-2.744, 1.824]	[-27.442, 18.237]

Note: Results for Within-Household model. Percentage changes in poverty indices for a 0.1 and 1 standard deviation increase in nightlight. 95% confidence intervals in brackets.

Appendix B: Figures

Figure B1. Nightlight Intensity in Vietnam (2012–2016)

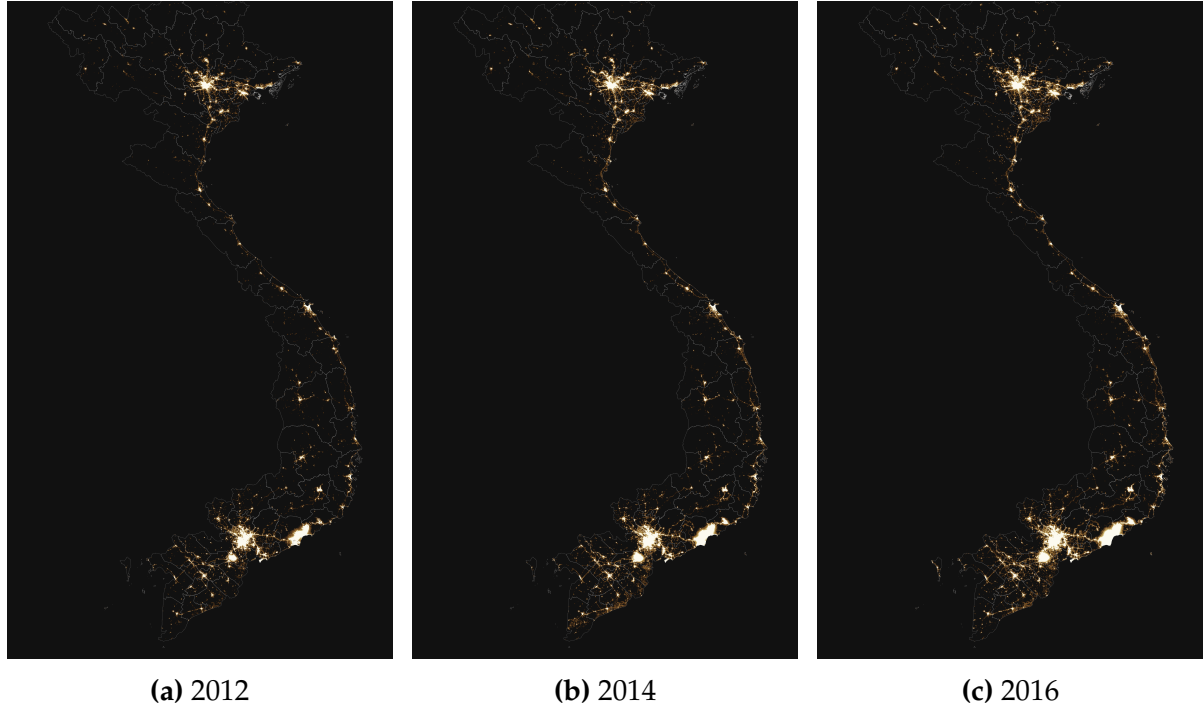
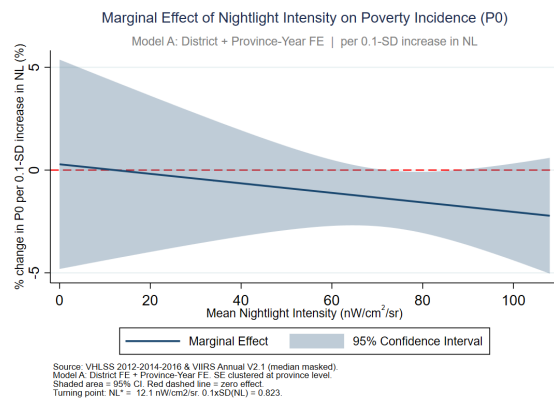
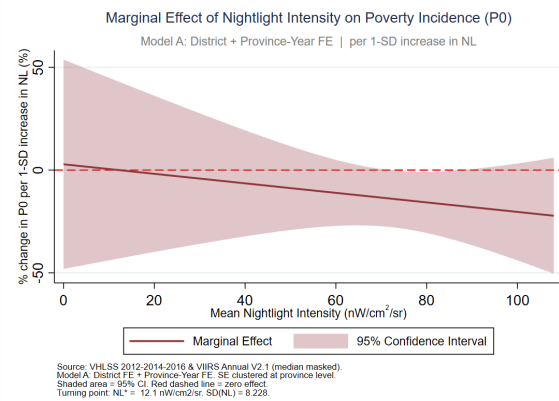
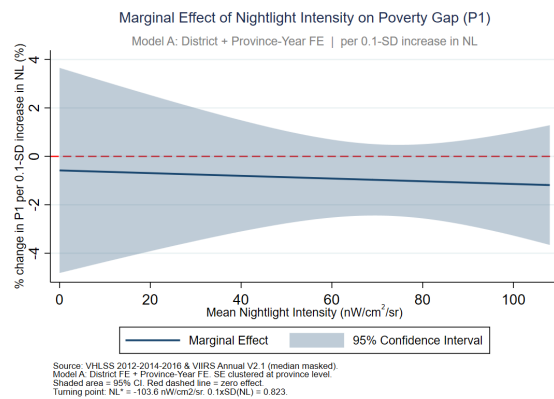


Figure B2. District-Level Predictive Margins for Poverty Incidence (P_0)

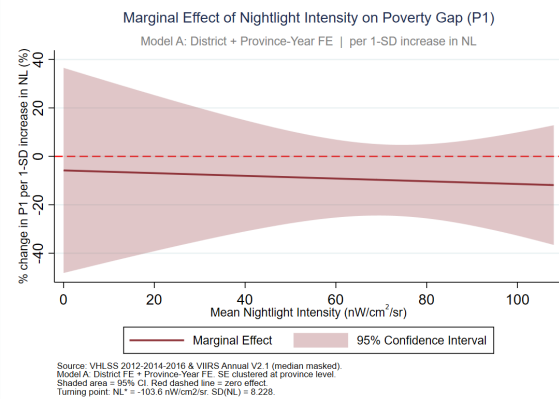
(a) Between-District / 0.1 SD



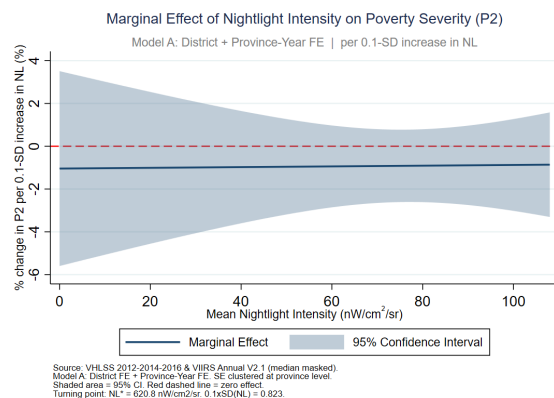
(b) Between-District / 1 SD

Figure B3. District-Level Predictive Margins for Poverty Gap (P_1)

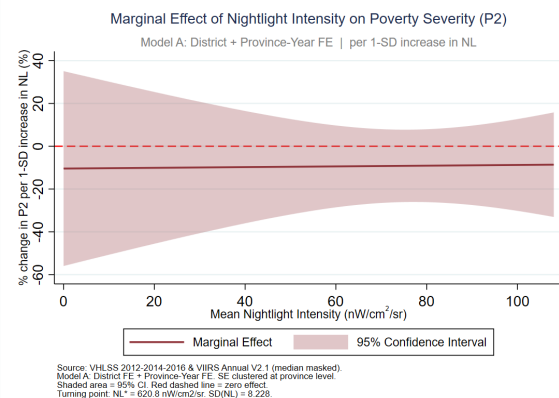
(a) Between-District / 0.1 SD



(b) Between-District / 1 SD

Figure B4. District-Level Predictive Margins for Poverty Severity (P_2)

(a) Between-District / 0.1 SD



(b) Between-District / 1 SD

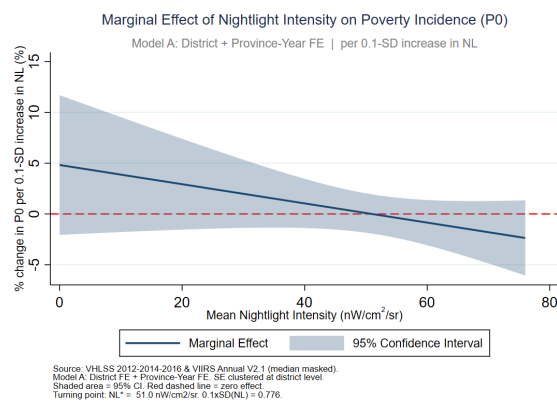
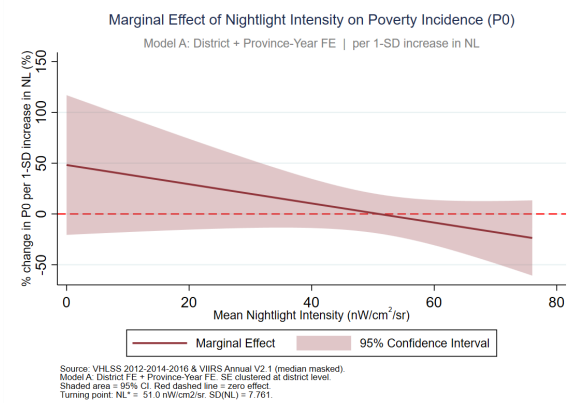
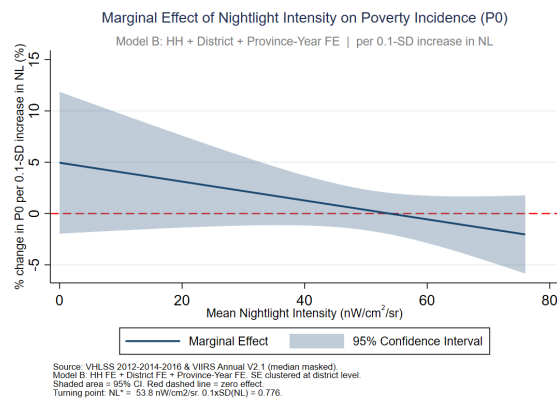
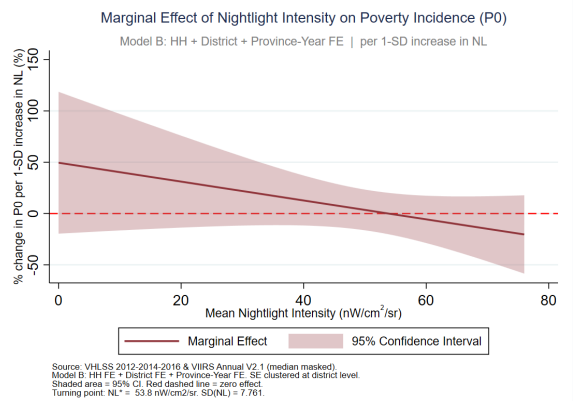
Figure B5. Household-Level Predictive Margins for Poverty Incidence (P_0)**(a) Between-District / 0.1 SD****(b) Between-District / 1 SD****(c) Within-Household / 0.1 SD****(d) Within-Household / 1 SD**

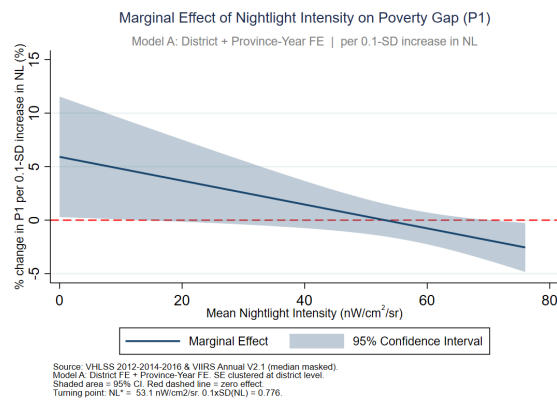
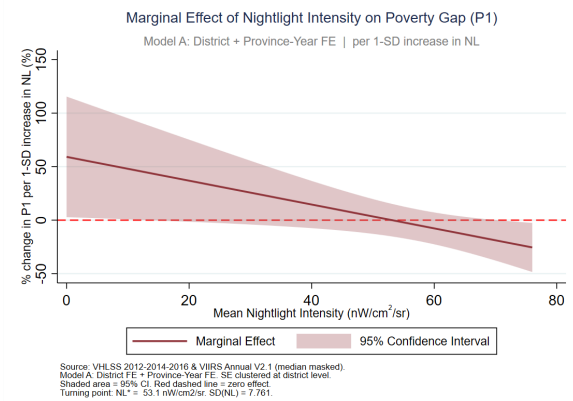
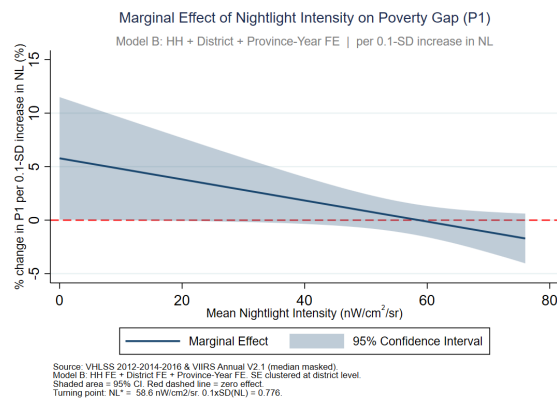
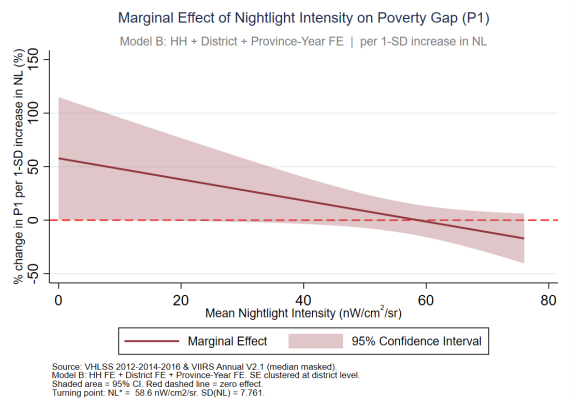
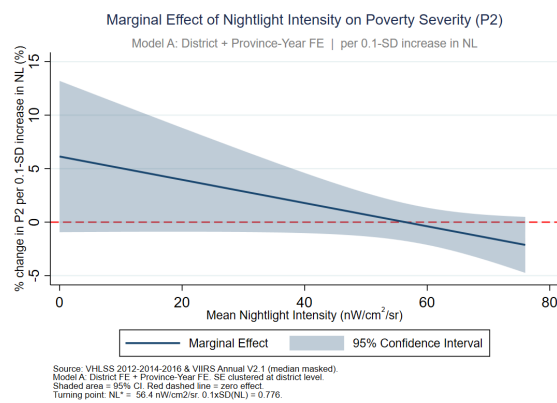
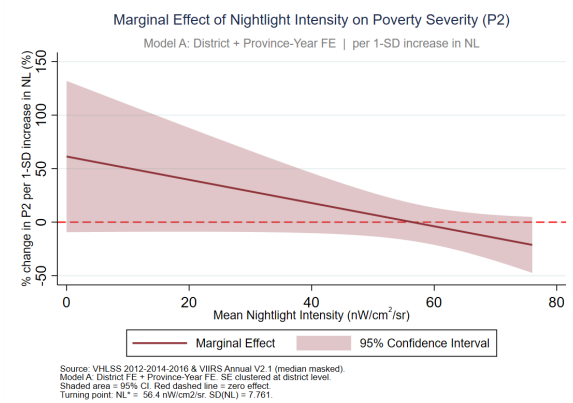
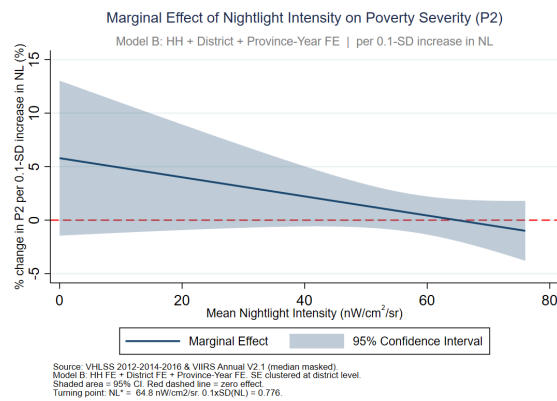
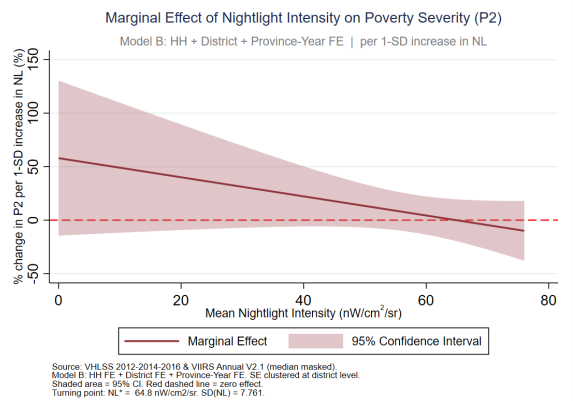
Figure B6. Household-Level Predictive Margins for Poverty Gap (P_1)**(a) Between-District / 0.1 SD****(b) Between-District / 1 SD****(c) Within-Household / 0.1 SD****(d) Within-Household / 1 SD**

Figure B7. Household-Level Predictive Margins for Poverty Severity (P_2)**(a) Between-District / 0.1 SD****(b) Between-District / 1 SD****(c) Within-Household / 0.1 SD****(d) Within-Household / 1 SD**